

Machine Tool Technology

Week 02

- 1 Lubrication- Need of Lubrication- Selection of Lubricant as per ISO standard.
2. Theory of metal Removal-Traditional & Non Traditional material removal process-Chip forming & Non chip forming
3. Types of cutting tools- Single point tools-Multi point tools -Specification of Single Point Cutting Tool, Parting, Knurling Tool, V-tool

Lubrication:

In a machine, lubrication is applying oil or grease on moving parts to help them move smoothly, reduce friction, prevent damage and keep the machine running well for a longer time.

Need of Lubrication:

- Reduce Friction
- Prevent Wear
- Cool Down
- Stop Rust
- Run Efficiently
- Last Longer
- Quiet Opera
- Stay Clean

Selection of Lubricant as per ISO standard:

Following are considered while selecting Lubricant as per ISO Standard:

i. Viscosity Grade:

Choose based on temperature and speed.

Common grades: ISO VG 32, 46, 68, etc.

ii. Type of Lubricant:

Mineral Oils: For general use.

Synthetic Oils: For extreme conditions.

Greases: For areas where liquid oils can't be used.

iii. Additives:

Anti-Wear (AW): For reducing wear.

Extreme Pressure (EP): For high-pressure conditions.

Rust Inhibitors: To prevent rust and corrosion.

iv. Operating Environment:

Consider temperature, humidity, and contamination. Choose lubricants suited to these conditions.

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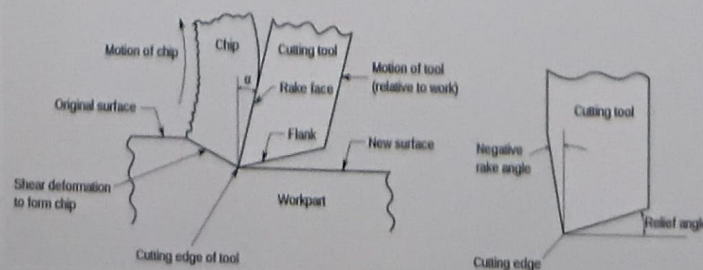
v. **Original Equipment Manufacturer [OEM]** Recommends Follow the machine manufacturer's guidelines.

vi. ISO Standards:

Refer to ISO 3448 for viscosity and ISO 6743 classification

Theory of metal Removal:

The Theory of Metal Removal explains how cutting tools interact with metal to shape it by forming chips, managing cutting forces, minimizing tool wear and controlling heat generation to achieve desired surface finishes and dimensional accuracy in machining processes.



The theory of metal removal is a fundamental concept in machining and manufacturing.

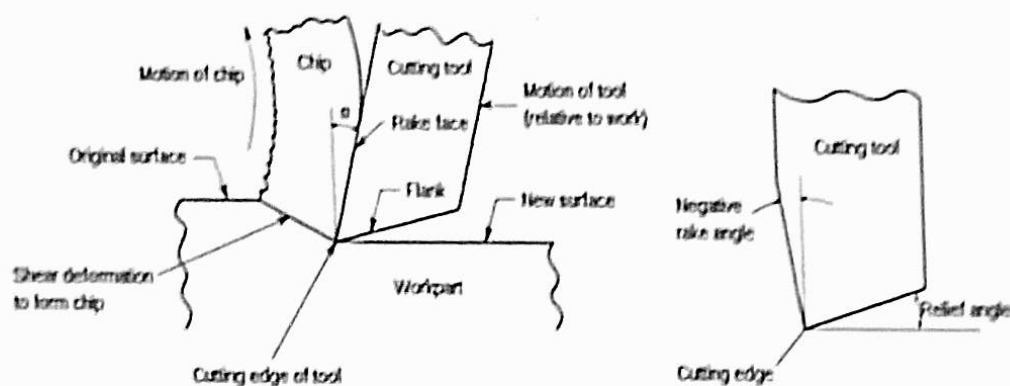
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The theory of metal removal is a fundamental concept in machining and manufacturing processes that explains how material is removed from a workpiece to achieve the desired shape, size, and surface finish.

Difference between Traditional & Non Traditional material removal processes:

Chip-Forming Machining Processes:

1. Turning: Rotates and shapes cylindrical parts using a cutting tool.
2. Milling: Uses rotating tools to cut flat or contoured surfaces from a workpiece.
3. Drilling: Creates holes in materials using a rotating drill bit.
4. Grinding: smooths surfaces using abrasive particles to achieve precision.
5. Broaching: Pulls or pushes a tool across a surface to cut complex shapes or keyways.

Non-Chip Forming Machining Process

1. Electrical Discharge Machining (EDM): Uses electrical sparks to shape materials, ideal for complex designs.
2. Laser Cutting: Uses a high-powered laser to cut materials with precision.
3. Water jet Cutting: Cuts materials using a high-pressure jet of water, ideal for various materials.
4. Ultrasonic Machining: Uses vibrations and abrasives to cut hard materials precisely.
5. Chemical Machining: Uses chemicals to shape materials, especially for delicate parts.

Types of cutting tools:

The cutting tools are as follows

- a) Single point cutting tool
- b) Double point cutting tool
- c) Multi point cutting tool

a) Single point cutting tool:

A single-point cutting tool is a tool that has only one cutting edge.

It is used in machining operations like turning, shaping, and planing where the tool removes material from the workpiece to shape it.



Advantages of single-point cutting tool:

- (i) Simple Design and easy to use.
- (ii) Cost-Effective, cheaper than multi-point tools.
- (iii) Can perform various tasks like turning and facing
- (iv) Easy to maintain,
- (v) Easy to control.

Disadvantages of single-point cutting tool:

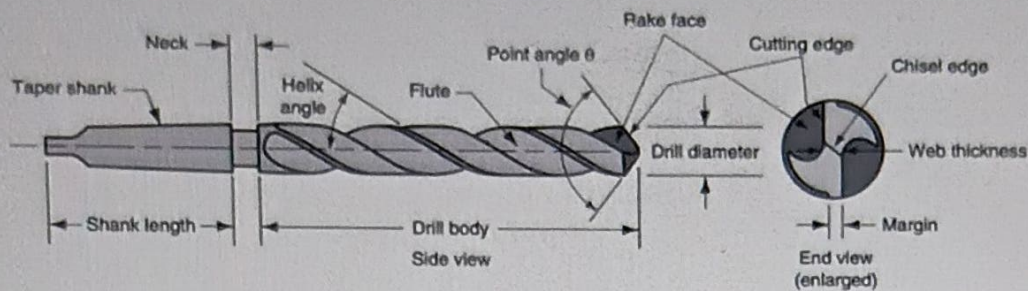
- (i) Removes material more slowly than multi-point tools.
- (ii) Tool wear out fast.
- (ii) May not achieve fine surface finish.
- (iv) Generate lot of heat which will effect tool life and work quality.
- (v) Less stable and causes vibrations.

b) Double point cutting tool:

A double-point cutting tool is a cutting tool that has two cutting edges used simultaneously in the machining process.

Example twist drill..

This too is commonly used in drilling operation.



Advantages of Double point cutting tool

- (i) More efficient, cuts more material in one pass.
- (ii) Reduces stress on each cutting edge.
- (iii) Provides a smoother surface finish.
- (iv) Less vibration during cutting.
- (v) Longer Tool Life.

Disadvantages of Double point cutting tool:

- (i) Complex Design
- (ii) Higher Cost.
- (iii) Need proper alignment before use.
- (iv) Not suitable for all machining operation.
- (v) Difficult to Maintain.

c) Multi point cutting tool:

A multi-point cutting tool is a tool that has multiple cutting edges, which simultaneously engage the workpiece during machining. Examples include drills, milling cutters, and broaches. These tools are commonly used in operations like drilling, milling and broaching to remove material more efficiently than single-point or double-point cutting tools.



Advantages Multi-point cutting tool:

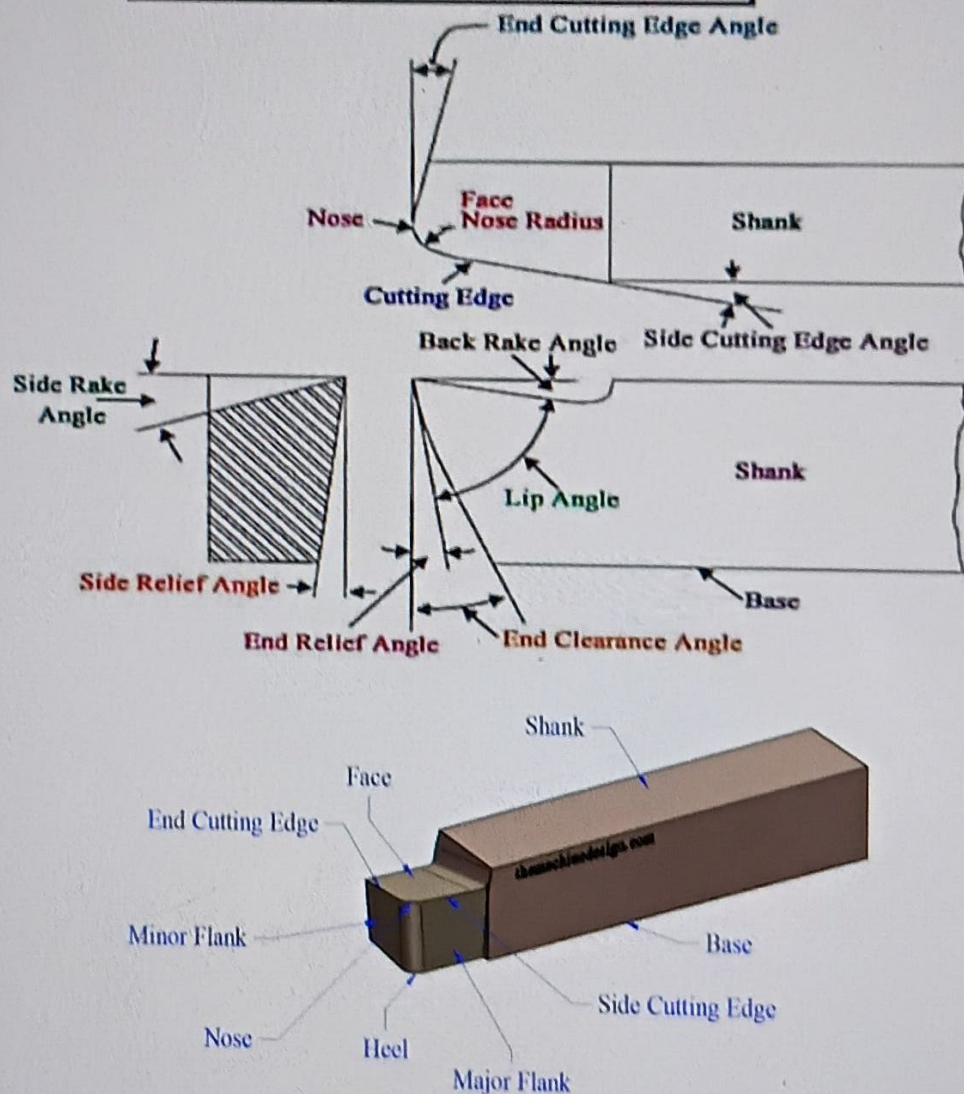
- (1) Removes material quickly.
- (ii) Provides a better surface finish.
- (i) Durable.
- (iv) Less Stress during operation.
- (v) High working speed.
- (vi) Can be used for various machining tasks.

Disadvantages Multi-point cutting tool:

- (i) Complex Design.
- (ii) Expensive.
- (iii) Difficult to maintain.
- (iv) Generates a lot of heat, needing good cooling.
- (v) Needs accurate alignment for best performance.

Specification of Single Point Cutting Tool

Single Point Cutting tool Diagram



Materials:

- High-Speed Steel (HSS)
- Tungsten Carbide
- Ceramics
- Diamond

Tool Geometry:

- **Shank:** The main body of the tool that is clamped
- **Cutting edge:** The part of the tool that removes material from the workpiece
- **Face:** The top surface of the tool where chips slide after cutting
- **Flank:** The vertical surface next to the cutting edge
- **Base:** The bottom surface of the tool
- **Nose:** The intersection of the major and minor cutting edges
- **Heel:** The curved area where the base and flank meet

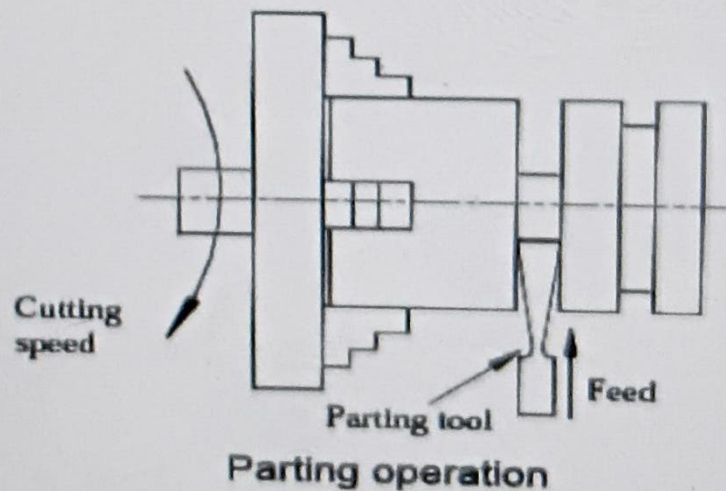
Tool Angles:

- **Rake Angle:** Angle between the tool face and a line perpendicular to the workpiece surface.
- **Clearance Angle:** Angle between the tool flank and the workpiece surface.
- **Side Cutting Edge Angle:** Angle between the side cutting edge and the workpiece surface.
- **End Cutting Edge Angle:** Angle between the end cutting edge and the workpiece surface.
- **Nose Radius:** Radius at the tool tip.

Uses:

Single point cutting tools are used to perform turning, facing, and producing flat surfaces on lathes, shapers, and planer machines.

Specification of Parting Tool



Materials:

- High-Speed Steel (HSS)
- Tungsten Carbide
- Ceramics
- Diamond

Dimensions:

Length: Total length of the tool.

Width: Width of the cutting edge

Height: Height of the tool shank

Thickness: Thickness of the cutting blade.

Tool Geometry:

Rake Angle: Angle between the tool face and a line perpendicular to the workpiece surface.

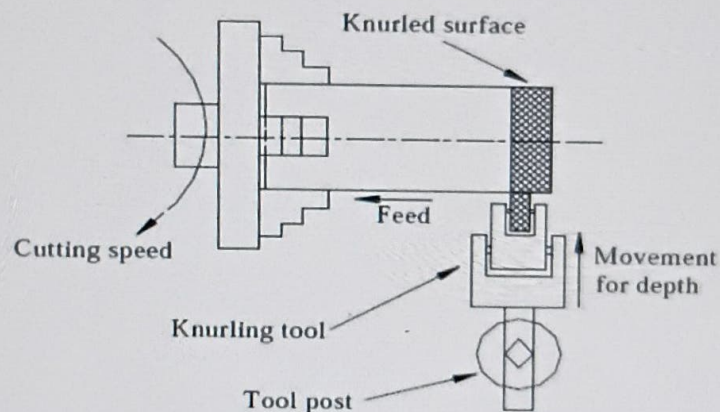
Clearance Angle: Angle between the tool flank and the workpiece surface.

Side Relief Angle: Angle to prevent the tool from rubbing against the workpiece.

Cutting Edge:

Sharpness: Degree of sharpness at the cutting edge

Nose Radius: Radius at the tip of the cutting tool to achieve a sharp edge. usually very small or

Specification of Knurling Tool**Dimensions:**

Length: Total length of the tool.

Width: Width of the tool shank.

Height: Height of the tool shank

Diameter of Knurling Wheels: Size of the knurling wheels.

Knurling Wheels:

Pattern Type: Straight, diagonal, or diamond.

Pitch: Distance between the peaks of the pattern (coarse, medium, or fine).

Number of Wheels: Typically 2 or 3 wheels.

Shank Size:

Width: Width of the shank.

Height: Height of the shank.

Wheel Mounting:

Fixed or Rotating Whether the wheels are fixed or can rotate